

HS Harz FuSa, Friedrichstraße 57

Airbag ECU

Messung und Verarbeitung der Sensoreingänge zur
Auslösung des Airbags
Measurement and Processing of Sensor Signals to Process
an Airbag Operation

Phase 10: Software Safety Requirements Version: 1.00

Contents

1	Meetings within the project.....	3
2	Unresolved topics.....	4
3	Glossar	5
4	Software Safety Requirements	6
4.1	Safety Relevant Functionalities	6
4.2	Software Requirements (SWR-01 to SWR-12).....	7
4.3	Crash Detection Logic and Thresholds	7
4.4	Safe State Definition	8
4.5	ASIL Allocation for Software Functions	8
5	Figure and table listing.....	9
6	List of dependent documents	10
7	Literature	11
8	Version history	12

1 Meetings within the project

Topic	Time	Room	Participants
Item Definition	13:00	Zoom	ALL
Bericht:	<u>none</u>		

2 Unresolved topics

No unresolved topics

3 Glossar

Tabelle 1: Glossar

Begriff / Term	Erklärung / Explanation
ACU	Airbag Control Unit.
ADC	Analog to digital converter value read from an analog sensor input.
ASIL	Automotive Safety Integrity Level used by ISO 26262 for risk classification.
CAN	Controller Area Network, a vehicle communication bus.
ECU	Electronic Control Unit.
EUC	Equipment under Control.
FuSa	Functional Safety.
HW	Hardware.
QM	Quality Management level below ASIL.
SIL	Safety Integrity Level.
SW	Software.
ACU	Airbag Control Unit.
ADC	Analog to digital converter value read from an analog sensor input.
ASIL	Automotive Safety Integrity Level used by ISO 26262 for risk classification.
CAN	Controller Area Network, a vehicle communication bus.
ECU	Electronic Control Unit.
EUC	Equipment under Control.
FuSa	Functional Safety.
HW	Hardware.

4 Software Safety Requirements

The software implementation of the Airbag ECU is responsible for processing crash sensor information, validating crash conditions, and activating deployment outputs under strict functional safety requirements. The implemented software logic follows the system architecture and safety requirements defined in the architecture and HARA documents.

4.1 Safety Relevant Functionalities

Tabelle 2 Safety Relevant Functionalities

ID	Function	Safety Relevance
F1	Crash detection	Main decision input for deployment.
F2	Airbag unfolding / output activation	Simulated deployment outputs.
F3	Sensor monitoring and plausibility check	Detects invalid sensor behaviour.
F4	System self-test and diagnosis	Detects startup and runtime failures.
F5	Fail-safe / fall-back state	Prevents deployment in detected fault state.
F6	Communication of safety relevant states	CAN/diagnostic communication of crash and fault status.

4.2 Software Requirements (SWR-01 to SWR-12)

Tabelle 3 Software Requirements

Requirement ID	Software Requirement	Purpose
SWR-01	The software shall read acceleration sensor input.	Crash detection
SWR-02	The software shall read gyroscope angular sensor input.	Rollover/angular motion detection
SWR-03	The software shall read pressure sensor input.	Side-impact detection
SWR-04	The software shall read passenger seat digital input.	Passenger airbag suppression
SWR-05	The software shall activate driver airbag output when valid crash is detected.	Driver protection
SWR-06	The software shall activate passenger airbag only if no baby seat is detected.	Passenger safety
SWR-07	The software shall activate warning lamp during system fault.	Fault indication
SWR-08	The software shall send crash/fault status through CAN.	ECU communication
SWR-09	The software shall read CAN messages for diagnostic or reset command.	Communication handling
SWR-10	The software shall read back output status after activation.	Output verification
SWR-11	The software shall enter safe state if fault is detected.	Prevent unintended deployment
SWR-12	The software shall support test routines for module and integration testing.	Verification

4.3 Crash Detection Logic and Thresholds

Tabelle 4 Crash Detection Thresholds

Sensor	Normal Range	Crash Threshold
Acceleration sensor	0-699	≥ 700
Gyroscope sensor	0-599	≥ 600
Pressure sensor	0-649	≥ 650

Frontal crash = acceleration sensor ≥ 700 .

Side crash = pressure sensor ≥ 650 .

Safing condition = gyroscope sensor ≥ 600 OR acceleration sensor ≥ 700 .

Deployment is allowed when: Crash Detected = TRUE AND Safing Condition = TRUE AND System Fault = FALSE.

Passenger airbag deployment additionally requires: Baby Seat Detected = FALSE.

4.4 Safe State Definition

The system enters safe state if: sensor value is outside valid range; output readback does not match the command; CAN error is detected; invalid passenger seat signal is detected.

Safe state is defined as: Driver airbag output = OFF, Passenger airbag output = OFF, Warning lamp = ON, CAN fault message = SENT.

4.5 ASIL Allocation for Software Functions

Tabelle 5 Tabelle 5 ASIL Allocation for Software Functions

ASIL	Prototype Function	Reason for Allocation	Related Requirement
ASIL D	Crash detection and deployment decision	A missed deployment or unintended deployment can be a severe safety hazard. Therefore the decision logic has the highest safety relevance in this prototype.	SWR-01, SWR-02, SWR-03, SWR-05, SWR-10, SWR-11
ASIL C	Passenger airbag suppression	Passenger airbag must be suppressed when a baby seat is detected, because wrong activation can create a serious passenger safety hazard.	SWR-04, SWR-06
ASIL B	Sensor plausibility, output readback and safe state	Fault detection prevents the ECU from operating with unreliable sensor or output information.	SWR-07, SWR-10, SWR-11
ASIL A	Warning lamp and diagnostic/CAN status	Fault and crash information must be indicated and communicated, but these functions do not directly command deployment.	SWR-07, SWR-08, SWR-09
QM	Serial debug print and university demonstration output	Useful for testing and explanation only; not part of the safety decision.	SWR-12

5 Figure and table listing

Tabelle 1: Glossar	5
Tabelle 2 Safety Relevant Functionalities.....	6
Tabelle 3 Software Requirements	7
Tabelle 4 Crash Detection Thresholds	7
Tabelle 5 Tabelle 5 ASIL Allocation for Software Functions	8

No table of figures entries found.

6 List of dependent documents

No.	Dokument name	Version	Date	File name
1	Arduino Coding and Logic Implementation for Airbag ECU System	V1	02.06.2026	FuSa Report_Adruido.docx
2	Airbag ECU ASIL Code Division	V02	02.06.2026	Airbag_ECU_ASIL_Code_Division_V02.docx
3	Hazard Analysis and ASIL Determination for Airbag ECU	V02	02.06.2026	Hazard Analysis and ASIL Determination for Airbag ECU V02.pdf

7 Literature

No.	Reference
[1]	ISO, ISO 26262-1:2011 Road vehicles - Functional safety. ISO 26262 addresses hazards caused by malfunctioning behaviour of E/E safety-related systems in road vehicles.
[2]	Synopsys, What is ASIL (Automotive Safety Integrity Level)? ASIL is a risk classification system under ISO 26262; ASIL A is lowest and ASIL D is highest.
[3]	Arduino Documentation, Due. Arduino Due is based on a 32-bit ARM core and provides 54 digital I/O pins, 12 analog inputs and CAN capability.
[4]	Arduino Documentation, Controller Area Network (CAN) Bus. CAN is a robust vehicle bus standard for communication between microcontrollers/devices without a host computer.
[5]	Atashi Saha, Arduino Coding and Logic Implementation for Airbag ECU System, uploaded project report, 2026.
[6]	HS Harz FuSa, Airbag ECU Portfolio HS Harz FuSa - Concept template, version 1.00, 2026/04/29.

8 Version history

Document name	date	Change log
Airbag_ECU_Phase_10	14.06.2026	v1.00 Initial version. Software Safety Requirements compiled from FuSa Report Sections 2, 6, 7 and ASIL Code Division V2 Sections 5.2, 5.3.